

CURRICULUM VITAE

NAME: THU NGOC MINH PHAN

POSITION TITLE: MSc Student

EDUCATION

Institution and Location	Degree	Completion Date	Field of Study
International University, Vietnam National University, Ho Chi Minh City (HCMC VNU), Vietnam	M.S.	2022-2025	Biomedical Engineering
University of Science, Vietnam National University, Ho Chi Minh City (HCMC VNU), Vietnam	B.S.	2017-2021	Medical Biotechnology

A. Personal Statement and Research Interest

I am a Master's student of Biomedical Engineering at International University, Vietnam National University HCMC. Being a dedicated researcher is the greatest dream of my life. I aspire to contribute significantly to advancing diagnostic and therapeutic methodologies to enhance human health. Thus, I always maintain a positive attitude, fostering my readiness to absorb new knowledge and skills. Working at Brain Health Lab, my research focused on EEG signal processing and data analysis to gain insights into brain activity, specifically targeting interventions for individuals experiencing mild cognitive impairment to prevent the progression to dementia in the early stages. In the future, I look forward to learning about exploiting the potential of artificial intelligence (AI) in neuro-science and neuro-engineering for clinical use.

B. Current and Past Positions

(02/2022 – present) Research assistant at Brain Health Lab, School of Biomedical Engineering, International University, Ho Chi Minh City - Vietnam National University HCM City, Vietnam.

- Main research direction: Focusing on early intervention methods for Alzheimer's disease, with Master's thesis titled “Effects of cognitive training games in Brain Train application on Mild Cognitive Impairment patients and applying Machine Learning models to predict intervention effect.”
- Participating in other lab projects, including Brain-computer interface (02/2022 - 02/2024), Depression and anxiety studies (06/2023 - 02/2024).
- (2023) Deputy organizing committee of the BrainConnects 2023 conference with the theme “Healthy Brain Aging: from Technology to Intervention.”
- Supporting my supervisor in guiding students in scientific research projects for academic courses (Project 1, Project 2, Pre-thesis).
- Supporting my supervisor (as teaching assistant) for Master's class in “Research Methodology” (Semester 1, 2023-2024), for PhD class in “Biostatistics” (Semester 2, 2024-2025).
- Being provided opportunities to fully develop scientific capabilities, including writing research proposals, leading projects, conducting experiments, performing analysis and interpretation, publishing research findings, preparing posters and presentations for participation in national and international conferences, attending training programs, seminars, webinars, and workshops.

(06/2020 – 10/2021) Trainee at Stem Cell Institute, University of Science, Ho Chi Minh City - Vietnam National University HCM City, Vietnam.

- Conduct Bachelor's thesis with the topic “Evaluation of the immunomodulatory properties and cartilage differentiation capability of umbilical cord mesenchymal stem cells in inflammatory microenvironment in vitro.”

- Cultivating and expanding mesenchymal stem cells (MSCs), and establishing co-culture systems for inflammatory studies.
- Differentiating MSCs into bone, cartilage, and fat cells, and analyzing cell morphology.
- Measuring cytokine levels and performing qPCR analysis to evaluate cartilage expression genes.

C. Awards and Honors

(2025) Second Prize of the 5th Vietnam Medical Achievement Award, Vietnam.

(2024) First Prize, Scientific Conference “Application of Digital Transformation and Artificial Intelligence in Disease Diagnosis and Treatment”, Vietnam.

(2024) IBRO-APRC Associate School on “Neuro-Enhancement for Brain Health”, Department of Physiology, Faculty of Medicine Siriraj Hospital, Mahidol University, Bangkok, Thailand.

(2023) Research Assistant Fellowship, School of Biomedical Engineering, International University, Ho Chi Minh City - Vietnam National University, Ho Chi Minh City, Vietnam.

(2023) Round 4 and the Grand Final Week, Social Business Creation, Montreal, Canada.

(2023) Third Prize in the contest “Digital Transformation - Creativity Challenge”, Ho Chi Minh City, Vietnam.

(2023) ADAI Fellowship, Ho Chi Minh City, Vietnam.

(2022) ITR Graduate Fellowship, Ho Chi Minh City, Vietnam.

D. Publications, Presentations and Posters

Publications:

Assessing the feasibility of cognitive intervention via BrainTrain cognitive gaming application for Mild Cognitive Impairment patients (Accepted, 2025, International

Journal of Bioinformatics Research and Applications). Authors: **Thu Ngoc Minh Phan**, Binh Nguyen Le Nguyen, Bao Quoc Nguyen, Nhung Hoang Do, Nhu Thi Huynh Tran, Nhan Thanh Nguyen, Thu Hoang Anh Pham, Nhan Quoc Trung Nguyen, Tuong Huu Nguyen, Kien Trung Duong, Nghia Tien Trong Hoang, Thu Thi Hoai Tran, Dang Minh Ly, and Dieu Xuan Nguyen Tan Le-Duy, and Huong Ha Thi Thanh.

Nguyen, M. T. D., Xuan, N. Y. P., Pham, B. M., Do, H. T. M., **Phan, T. N. M.**, Nguyen, Q. T. T., ... & Ha, H. T. T. (2023). Optimize temporal configuration for motor imagery-based multiclass performance and its relationship with subject-specific frequency. *Informatics in Medicine Unlocked*, 36, 101141.

Truong, N. C., **Phan, T. N. M.**, Huynh, N. T., Pham, K. D., & Van Pham, P. (2023, June). Interferon-Gamma Increases the Immune Modulation of Umbilical Cord-Derived Mesenchymal Stem Cells but Decreases Their Chondrogenic Potential. In *International Conference Innovations in Cancer Research and Regenerative Medicine* (pp. 19-33). Cham: Springer International Publishing.

Phan, T. N. M., Nguyen, H. T., Huynh, T. N. M., Nguyen, T. H., Tran, T. N. Y., & Ha, H. T. T. (2022, December). Development and Commercialization of a Brain Training App Targeting the Vietnamese Elderly. In *International Conference on the Development of Biomedical Engineering in Vietnam* (pp. 719-737). Cham: Springer Nature Switzerland.

Presentations:

(2024) 14th Mind-Body Interface Internationals Symposium, China Medical University, Taichung, Taiwan: “Assessing the Feasibility of Cognitive Intervention via BrainTrain Cognitive Gaming Application for Mild Cognitive Impairment Patients.”

(2024) Scientific Conference “Application of Digital Transformation and Artificial Intelligence in Disease Diagnosis and Treatment”: “Development and testing of the BrainTrain cognitive training application for Mild Cognitive Impairment (MCI) patients.”

(2023) Hyper-Interdisciplinary Conference 2023 (Flash talk): “Development and Commercialization of a Brain Training App Targeting the Vietnamese Elderly.”

(2022) The 9th International Conference in Vietnam on the development of Biomedical Engineering: “Development and Commercialization of a Brain Training App Targeting the Vietnamese Elderly.”

Posters:

(2024) International Conference on Neuroscience and Brain Health 2024 (ICNB2024) and the 27th Thai Neuroscience Society Annual Conference (TNS27): “Assessing the Feasibility of Cognitive Intervention via BrainTrain Cognitive Gaming Application for Mild Cognitive Impairment Patients.”

(2024) Scientific Conference “Application of Digital Transformation and Artificial Intelligence in Disease Diagnosis and Treatment”: “Development and testing of the BrainTrain cognitive training application for Mild Cognitive Impairment (MCI) patients.”

(2021) BrainConnect Conference 2021: “Development and Commercialization of a Brain Training App Targeting the Vietnamese Elderly.”

E. References

Huong Thi-Thanh Ha, Ph.D.

Head of Brain Health Lab

Chair, Department of Tissue Engineering and Regenerative Medicine

School of Biomedical Engineering, International University

htthuong@hcmiu.edu.vn

Nhat Chau Truong, M.S.

Head of Center for Biomedical Training

Stem Cell Institute, University of Science

nhattruong@sci.edu.vn